

NIKHIL KUSHWAH

B.TECH, ELECTRICAL ENGINEERING SATI, VIDISHA

PROFESSIONAL SUMMARY

Electrical Engineering student passionate about practical learning and hands-on hardware design, with secondary expertise in software development and embedded coding to build end-to-end technical solutions.

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Ganj Basoda, Vidisha, (M.P.)

LinkedIn:

<https://github.com/nikhilkush078/>

EDUCATION

SAMRAT ASHOK TECHNOLOGICAL INSTITUTE, VIDISHA

B.Tech - Electrical Engineering - CGPA (8.2)

2023 - 2027

NAVANKUR VIDYAPEETH HSS, GANJ BASODA

HIGH SCHOOL (10TH) - 65%

HIGHER SECONDARY (12TH) - 78.8%

2021-2023

TECHNICAL SKILLS

Programming Languages: Python (Pygame, Tkinter- GUI, Opencv - Computer Vision) , HTML, CSS, JavaScript (Basics), Arduino Programming

Software: FreeCad, Arduino IDE, Photoshop, Canva, VScode, Tinkercad, Circuit Wizard, Proteus, Ms office , Linux OS, EassyEDA (PCB Designing)

Hardware: Arduino (Uno, Nano, Mega) , ESP32, ESP8266, SMPS and Electronics Circuits Repairing

Other Skills : YOLO-V8 (Machine Learning Modal), Computer Vision Using OpenCV

TECHNICAL CONTENT & OPEN SOURCE

GitHub: More than 35+ public GitHub repositories with complete project descriptions.

Linkdin: More than 20+ Complete R&D based projects demonstration videos

YouTube: More than 20+ OpenSource Project Videos, Live Electronics Repairing

PROJECTS

Campus Autonomous Navigation & Delivery Vehicle

Designed and deployed a computer vision-driven autonomous control framework leveraging OpenCV and custom-trained YOLO object recognition models for real-time obstacle perception and navigation. Developed the software architecture for a full-scale campus vehicle concept before deploying the complete vision stack onto a functional, medium-sized delivery robot to test real-world viability. Field-tested the scaled prototype across campus, validating successful point-to-point payload delivery and proving the reliability of the vision pipeline under real-world operating conditions.

Demonstration Video Link -

<https://youtu.be/hHUhdu7TY9Y>

Spark Detection and Automatic Power Cutoff System

Designed and implemented an intelligent electrical safety system that detects AC outlet arcing by analyzing high-frequency line distortion using an Arduino microcontroller. The system monitors power line signal signatures to detect transient spark noise in real time and automatically actuates a relay to trip the main power circuit, preventing high-current hazards and mitigating electrical fire risks.

Demonstration Video Link -

https://www.linkedin.com/posts/nikhil-kushwah-664304218_spark-detector-activity-7347909211177082880--w9T?utm_source=share&utm_medium=member_desktop&rcm=ACoAADbcwqWbmaWPICVZ41SWioXWKvjFBOHsi24

Fire Detection System using Computer Vision (YOLO + OpenCV + Python)

Designed and developed an intelligent fire-detection system using computer vision and machine learning. Trained a custom YOLO model to accurately identify fire in real-time video streams and integrated it with OpenCV for image processing. Implemented the system in Python, enabling fast detection and generating alerts when fire patterns were recognized. This project demonstrates skills in model training, dataset preparation, real-time inference, and practical computer-vision deployment.

Demonstration Video Link -

https://www.linkedin.com/posts/nikhil-kushwah-664304218_fire-detector-activity-7359463770810658818-OxFC?utm_source=share&utm_medium=member_desktop&rcm=ACoAADbcwgwBmaWPICVZ41SWioXWKvjFBQHsl24

Low-Cost Underground PVC Pipe Detection System

Conceptualized and built an ultra-low-cost utility tracking system specifically designed to map non-conductive (PVC) subterranean water pipes—solving a key limitation of standard metallic pipe locators. Designed a high-frequency RF signal injection module that utilizes the internal water column as a signal path via accessible outlets, coupled with a custom-tuned handheld RF receiver to trace radiated magnetic fields from above ground. Validated precise underground path mapping in field tests, delivering a high-reliability, low-cost alternative to industrial Ground Penetrating Radar (GPR) equipment.

Demonstration Video Link -

<https://lnkd.in/p/d2BMQ6Ji>

SOME OTHER PROJECTS

Human Body Detection System (Capacitive Coupling Method)

Built a non-contact sensing system that detects human presence using capacitive coupling principles, similar to a body-proximity detector.

Software-Controlled Interactive Robot (Python-Based Control)

Developed a robot capable of interacting with people—seeing, responding, and performing actions—controlled through a Python software interface.

Object-Finding Robot using YOLO and OpenCV

Created a mobile robot that detects and searches for a specific object using a custom-trained YOLO model for real-time object recognition.

Video demonstrations for all the projects below—along with several additional hardware and software builds—are available on my LinkedIn and YouTube profiles.

ACHIEVEMENTS

- **Smart India Hackathon 2024 Finalist**
- Got **First** Position in **Paper Presentation Competition** Organize by EE Dept in TechnoVision 2024
- Got **Second** Position in **Energy Efficient Project Competition** Organize by EE Dept in TechnoVision 2024
- Got **Second** Position in **Academics** (in EE branch) in 2024
- Got **Third** Position in **Academics** (in EE branch) in 2025

POSITION OF RESPONSIBILITY

- **Student Coordinator** of **AICTE IdeaLab** of our collage
- **IOT Head** in GDGOC(**Google Development Groups** On Campus) SATI Vidisha
- **Co-Convenor** Startup-Cell Sati Vidisha
- Ex - **Co-Convenor** and member of **Technical Club Flux** (official club of our college)

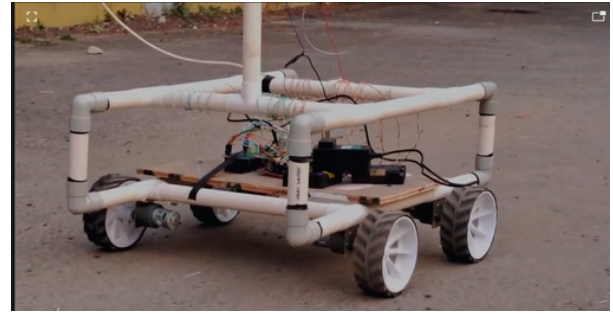
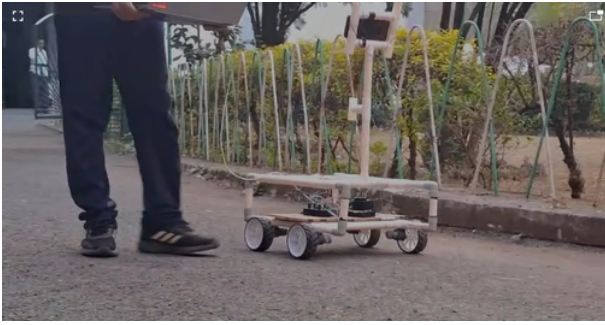
INTERSHIPS

- Inhouse Internship on **Robotics and Pcb Design** Organised by EE Dept 2024
- Inhouse Internship on **Embedded Systems** Organised by EE Dept 2025
- Industry internship at **Navgyan Constructions** on Mahawadiya-Khajuri 33kv Line Project 2026

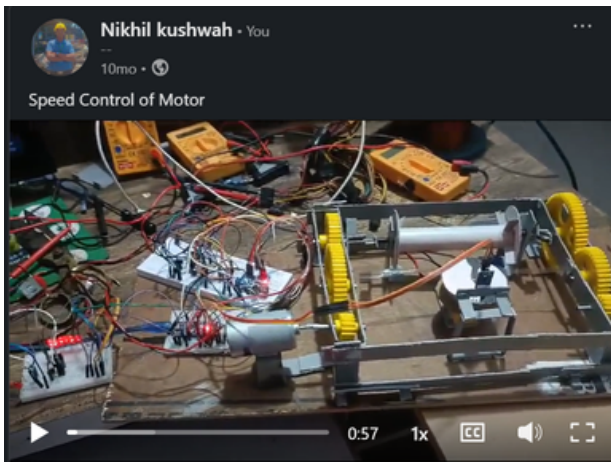
Connect With Me



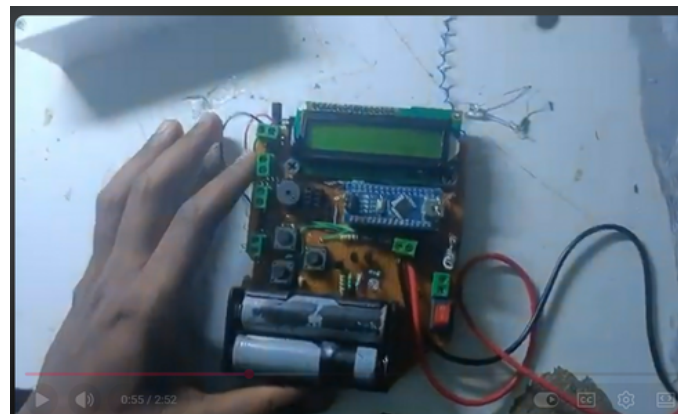
Hardware Projects Images



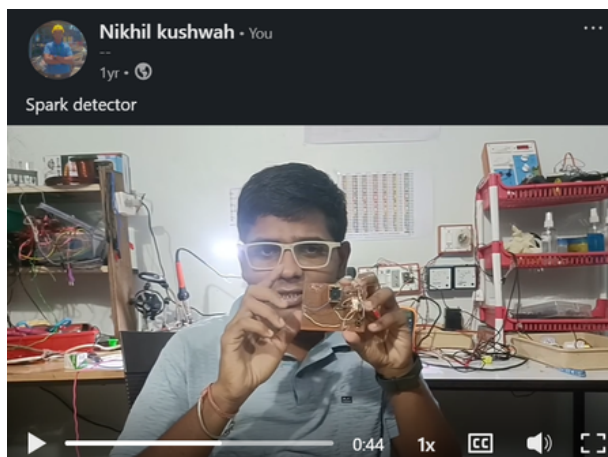
Self-Driving Delivery Bot



Speed Control Of DC Motor Using Pulse Width Modulation



Digital Multimeter And Component Tester



Spark Detection Circuit Breaker System



Electric Shock Protection System

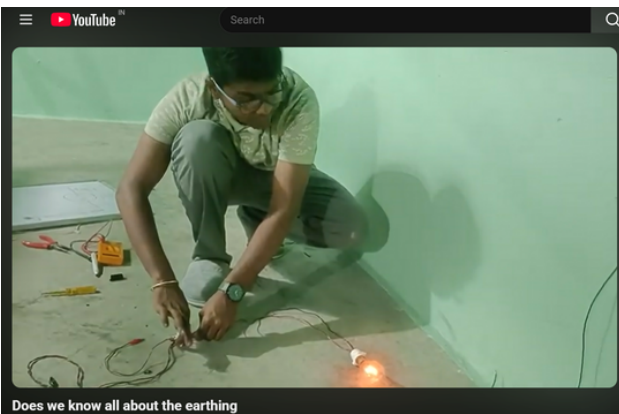
These photos capture my journey over the last three years—from assembling circuits on the workbench and debugging embedded code to presenting working engineering prototypes



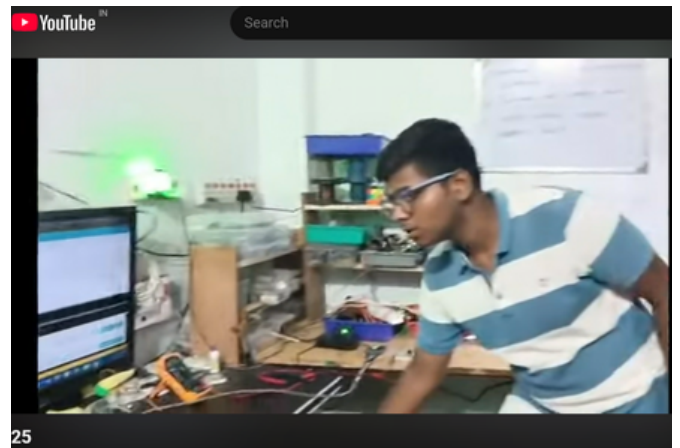
Hardware Projects Images



Self-Deriving Car For COLlege Campus (Project Sim-V)



Experiment On Pilar As Earthing



Computer Vision Based Intractive Game

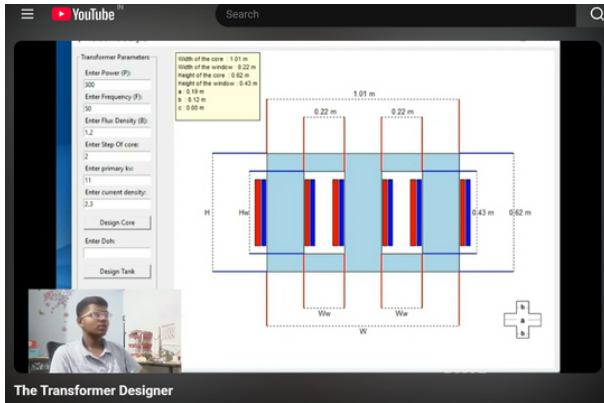


Smart India Hackthon Gaziabad 2024

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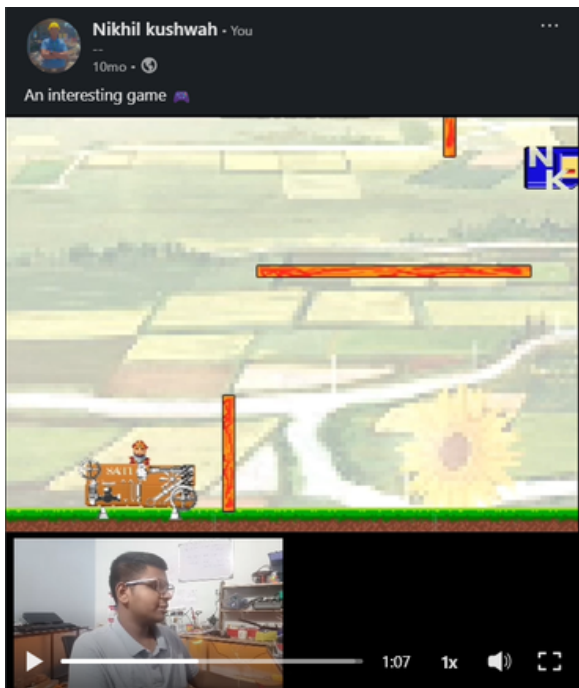
Software Projects Images



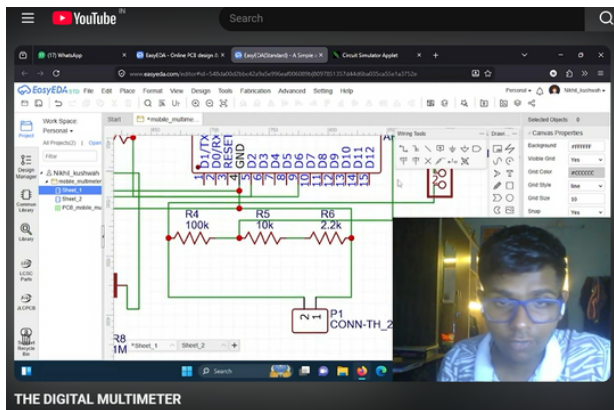
Transformer Designer Software
Made Using **Python - TinkerGUI**



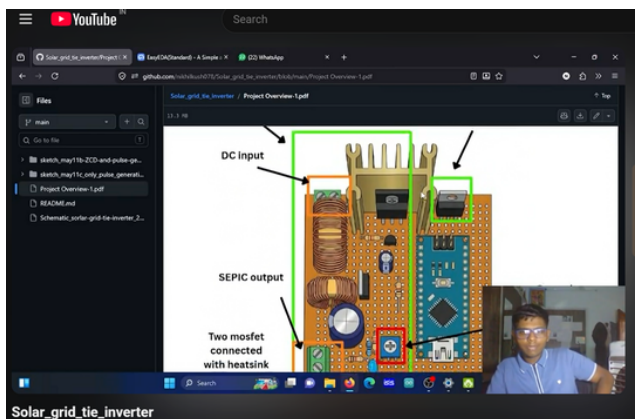
Thyristor Circuit Waveform Simulator
Made using **Python - Matplotlib**



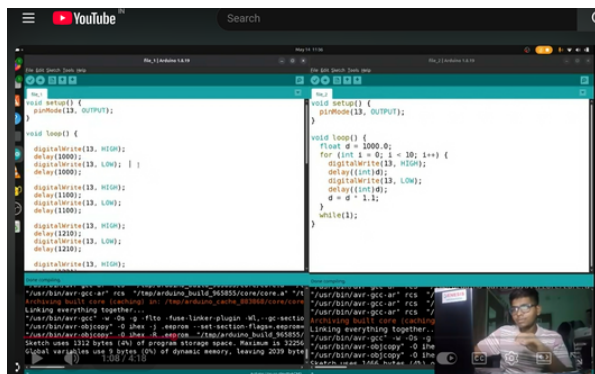
Game Development Using
Python - Pygame



Grid Tie Inverter Schmatic Design
Using **EasyEDA**



Desing And Development of 3D
PCB Designing Using **EasyEDA**



Arduino Programing and
Storage utilization

The screenshots and images above showcase software applications, GUI tools, and computer vision systems I have developed and demonstrated over the past three years. Scan the attached QR codes to view live software walk-throughs and repository code.

